

ASSIGNMENT 4

Aim

To design and develop a Flask-Powered Dynamic To-Do List Application that demonstrates full-stack web development using HTML5, CSS3, JavaScript, Python 3, and Flask.

The application allows users to Add, Edit, Delete, Mark Complete, Filter, and Prioritize Tasks with a Flask backend handling data through REST APIs

Tasks Performed

Project Setup & Flask Initialization

Flask REST API Routes

HTML Template & Layout

JavaScript & Fetch API Integration

CSS Styling & Responsive Design

Testing & Code Quality

TECK STACK

HTML, CSS, JS , PYTHON, FLASK

CODE:

```
app.py 2 X
app.py > ...
1 from flask import Flask, render_template, request, jsonify
2 from datetime import datetime
3
4 app = Flask(__name__)
5
6 tasks = []
7 next_id = 1
8
9
10 # Home Route
11 @app.route("/")
12 def home():
13     return render_template("index.html")
14
15
16 # GET All Tasks + Filter
17 @app.route("/api/tasks", methods=["GET"])
18 def get_tasks():
19
20     status = request.args.get("status")
21
22     if status == "active":
```

```
app.py 2 index.html X
templates > index.html > html
1 <!DOCTYPE html>
2 <html>
3 <head>
4
5 <title>Professional To Do App</title>
6
7 <link rel="stylesheet"
8 href="{{ url_for('static',
9 filename='css/style.css') }}">
10
11 </head>
12
13 <body>
14
15 <div class="container">
16
17 <h1>My To-Do List</h1>
18
19
20 <input
21 id="title"
22 type="text"
```

```
app.py 2 # style.css x
static > css > # style.css > " "
1  *{
2  margin:0;
3  padding:0;
4  box-sizing:border-box;
5  font-family:Segoe UI;
6  }
7
8  body{
9  min-height:100vh;
10 display:flex;
11 justify-content:center;
12 align-items:center;
13 background:
14 linear-gradient(
15 135deg,
16 #667eea,
17 #764ba2
18 );
19 padding:30px;
20 }
21
22 .container{
```

```
JS app.js x
static > js > JS app.js > ...
1 const API="/api/tasks";
2
3
4 async function loadTasks(){
5
6 let res=
7 await fetch(API);
8
9 let data=
10 await res.json();
11
12 showTasks(data);
13
14 }
15
16
17
18 async function addTask(){
19
20 let title=
21 document.getElementById(
22 "title"
```

Output:

